# Editorial

BT launched its public broadband service in 1996, and started the process which is likely to lead to the replacement of the existing digital network in the next millennium. Broadband, supporting single connections of bandwidths from 64 kbit/s upwards, provides the promise of a vast range of services integrated on to a single network. However, current implementations support only fixed point-to-point connections — permanent virtual channels — offering a simple digital service similar to the 'nailed up' circuit in narrowband digital switches. To broaden the service to any more than a handful of customers, on-demand connections — switched virtual channels — are vital.

To allow diverse types of communications to be transported, a new technology was needed. Asynchronous transfer mode provides the capability to support a range of different connections, from the variable bit rate, non-real-time requirements for data communications (such as Internet access), to the constant bit rate, real time demands of telephony and video. Broadband is perhaps a poor term for such a technological development; perhaps flexible band would be a better description.

The transportation of bits, involving transmission and switching technologies, does not by itself provide a service. Services are enabled by automated communication between the user's equipment and the network, and implemented throughout the network by communication within and between the networks themselves. This communication has evolved through the convergence of telecommunications, distributed processing, information networking and software disciplines, to produce a unique type of messaging for the control of calls and connections — signalling.

The control systems proposed for switches and networks also reflect the convergence of these technologies. The fluctuating, changing market of data networks (local area networks especially) demanded fast implementations and simpler protocols such as the ATM Forum's Private Network/Network Interface, described in the paper by Jennifer Scott and Ian Jones, p 37, and developed in a new form focused on ATM and overviewed in the paper by Nick Cooper, p 29. New markets are also being explored, with services such as video-on-demand being defined and refined in the Digital Audio-Visual Council (DAVIC), as explained in the paper by Richard Miles et al, p 87.

The complexity as well as the flexibility of broadband is reflected in the number of standards organisations, which are identified and detailed in the paper by Bryan Law, p 11. While the newer organisations looked at the hard reality of implementations based on current requirements, the world's public network operators, with their global networks, reliable connections and huge capital investment programmes, have their sights set a little further into the future. Global networks providing control structures that can be customised to ever-changing requirements were first propounded in the Telecommunications Information Network Architecture Consortium (TINA-C), a global organisation that included BT. Among the many resulting research projects was a new way of tackling signalling protocols — described in the paper by Paul McDonald, p 169 — and the separation of call and bearer control, described in the paper by Dick Knight and Bryan Law, p 75. The goal of a generic platform with globally distributed control structures remains firmly with the more traditional standards body, the International Telecommunications Unions (ITU), and insight into the signalling requirements for that goal can be found in the paper by Gary Bruce and Jon Clark, p 178. The ITU have already completed and published their first step along the road, with protocols to control primary connections — the paper by Bryan Law and Dick Knight, p 58, details the UNI protocol, the SS7-based interswitch control is explained in the paper by Ian Jones, p 47, and the paper by Mick Hale et al, p 99, describes the control of access networks using the VB5 interface.

The demand for traditional fixed telephony is being replaced by a requirement for voice services offering mobility and the same reliability and diverse services as the copper-based solutions. The solution leads towards a goal of fixed mobile integration — the third generation mobile systems, as outlined in the paper by Alan Clapton et al, p 120. UMTS will make almost no distinction between hand-held radio terminals and traditional landline connections for voice-based services.

Integrating all of the service requirements into control structures will provide customers with the services they require, on demand, at the bit rates they require and with the quality they require. Some of the diverse services that provide the signalling requirements are described in the paper by Roger Morley and Dick Knight, p 20, and the paper by Frank Allard, p 112. Signalling is the glue between the services and the transport, providing networking solutions for the next generation of switches, through standardised communications for distributed processing systems. While the theory has been laid down, mainly in standardisation bodies, it is always important to be able to implement the standards! BT Laboratories is world-renowned for its practical application of research and development, and the examples detailed in the papers by Chris Shephard et al, p 141, and Paul Reece et al, p 155, show some of the practical achievements in broadband signalling.

The performance of control systems has long been recognised as an issue, and some of the results of investigations are given in the paper by David Morris and Anne Elvidge, p 132.

The papers detailed above are mainly the work of people from the Network Intelligence Engineering Centre, a major source of broadband signalling expertise within BT's Systems Engineering department. The authors' experiences come from projects delivering services to standards activities looking far into the future, including practical experiences building demonstrators and bench models, and research projects like RACE MAGIC that have yet to reveal the full impact of their results. Close working with colleagues from a number of units, and also a number of notable world experts, who were all determined to achieve the highest quality for their papers, has made my task not only the easiest job of my career, but also one of the most enjoyable. I hope that you derive the same level of satisfaction as you read the material we have been able to present.

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*Guest Editor*